

packages/graph-view/src/utils/elk-bundled.ts

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Overview

A new file `packages/graph-view/src/utils/elk-bundled.ts` (R1-R35) adds a Vite/Electron-friendly shim for the ELK layout engine. It declares typed constructors, resolves the UMD bundle's export, and lazily loads the engine and its instance.

Key changes

- **Type definitions** (R3-R7):
 - `ElkLayoutInstance` with a `layout(graph: unknown): Promise<unknown>` method.
 - `ElkConstructor` as `new () => ElkLayoutInstance`.
- **resolveElkConstructor** (R9-R14): inspects the imported module for `default` or `ELK`; throws an error if neither is a function.
- **loadElkConstructor** (R19-R25): imports `elkjs/lib/elk.bundled.js` once, caches the constructor promise, and returns it.
- **loadElk** (R30-R35): creates a single `ElkLayoutInstance` from the cached constructor, caching the instance promise.
- **Shim comment** (R1): explains the need for this file in Vite/Electron contexts where the UMD bundle lacks an ESM default export.

Impact

- **Correctness**: ensures the ELK constructor is resolved from the UMD bundle, throwing a clear error if missing.
- **Performance**: caches both constructor and instance promises, avoiding repeated imports or instantiations.
- **Maintainability**: centralizes ELK loading logic; other modules can import `loadElk` without bundler quirks.
- **Compatibility**: works with Vite/Electron setups that otherwise cannot import the UMD bundle as an ES module.

Risks & follow-ups

- **Import path resolution**: verify that `elkjs/lib/elk.bundled.js` resolves correctly in all target environments.
- **Error handling**: ensure the thrown error message is surfaced to users when the bundle is missing or malformed.
- **Caching correctness**: test that repeated calls to `loadElk` return the same instance and that promise rejection propagates properly.

- **Type safety:** run TypeScript linting to confirm that the new types integrate cleanly with existing graph-view code.